Basics of Information Security (4360702)

Semester – 6th

Diploma in Computer Engineering

Prepared By:
Y.A.Hathaliya
Lecturer in Computer Engg. Department
A.V.P.T.I, Rajkot

Chapter 4
Network Security

Topics To be Covered

- Security Topologies
  - Security Zones, DMZ,
  - Internet, Intranet
  - VLAN, Security Implication in VLAN
  - Tunneling.
- Firewalls
  - Working of Firewall
  - Need of Firewall
  - Types of Firewall (Packet Filtering Firewall, Circuit-Level Gateway Firewall, Application-Level Gateway Firewall, Stateful Inspection Firewall, and Next-Generation Firewall)
- Intrusion detection systems (IDS)
  - Definition of Intruders
  - Components of IDS
  - Types of IDS with Advantages and Disadvantages (HIDS,NIDS)

Security Topologies

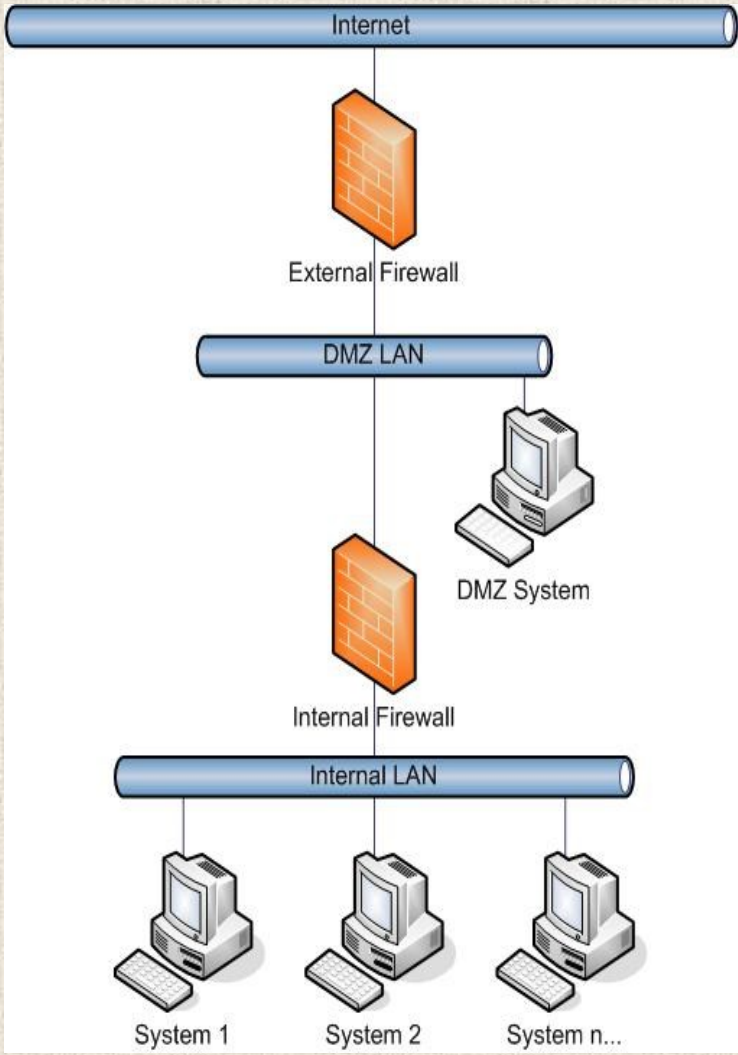
- Security Topologies refer to the architectural designs and configurations used to establish and maintain the security of computer networks and systems.
- These Topologies are crucial for protecting information, resources, and infrastructure from unauthorized access, data breaks, and other cyber threats.
- Different organizations may implement various security topologies based on their specific requirements, the nature of their operations, and the sensitivity of the data they handle.
- Example of Security Topologies are.
  - DMZ
  - Network Segmentation
  - VLAN (Virtualization Security)

Security Zones

- Security Zones in network security refer to the concept of dividing a network into distinct areas, each with its own set of security policies, controls, and levels of trust.
- This segmentation is designed to enhance overall security by limiting the scope and impact of security incidents and unauthorized access, each security zone is treated as a separate entity, and traffic between zones is subject to specific rules and scrutiny.
- Imagine security zones as rooms in your home, accessible only to specific family members. For instance, your home office may be restricted, barring entry for helpers and children, security zones help regulate entry to specific spaces and ensuring the confidentiality and security of sensitive information.
- Here are some common security zones in network architecture,
  - DMZ (Demilitarized Zone)
  - Remote Access Zone
  - Internal Zone
  - Guest Network Zone
  - VPN Zone

DMZ

- Demilitarized Zone (DMZ) is a network security concept that involves creating a separate, intermediate network segment between an organization's internal network (trusted network) and an external network (untrusted network), typically the internet.
- The purpose of a DMZ is to provide an additional layer of security by segregating public-facing services, such as web servers, email servers, or FTP servers, from the internal network where sensitive data is stored.
- In the context of cybersecurity, a DMZ acts as a buffer zone that helps protect internal resources from direct exposure to the internet, It often contains firewalls, intrusion detection/prevention systems, and other security appliances to monitor and control traffic entering and leaving the network.
- The basic figure of DMZ are depicted below,



- Characteristics/Application of DMZ are;
  - Segregation of Services
    - DMZ allows organizations to segregate their public-facing services, such as web servers, email servers, or FTP servers, from their internal network.
    - This separation helps in containing potential security breaches and limiting their impact to only the DMZ.
  - Enhanced Security
    - By placing critical services in the DMZ, organizations can implement stricter security measures for these services
  - Restricted Access
    - Access controls are typically implemented to restrict traffic flow between the DMZ and internal networks.
    - Only specific types of traffic necessary for the operation of public-facing services are allowed, while all other traffic is blocked or carefully scrutinized.
  - Web Servers
    - Hosting web servers in the DMZ allows external users to access a company's website without directly connecting to the internal network, safeguarding sensitive data.
  - Email Servers
    - Placing email servers in the DMZ ensures that external communications, such as emails from clients or partners, are processed in a secure environment before reaching the internal network.

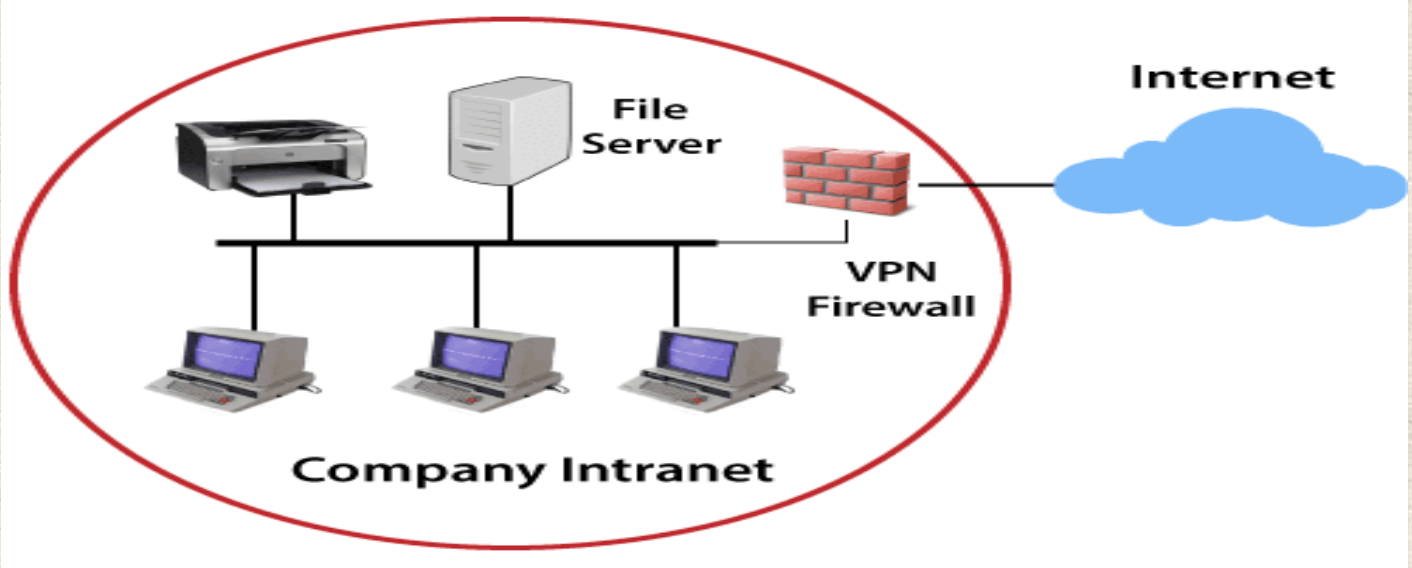
- FTP Servers
  - File Transfer Protocol (FTP) servers in the DMZ facilitate secure file transfers between internal and external networks.
- DNS Servers
  - Domain Name System (DNS) servers in the DMZ handle external DNS requests, separating this critical service from the internal network to prevent potential attacks.
- Authentication Services
  - Placing authentication services in the DMZ allows external users to authenticate without gaining direct access to internal user databases, enhancing security.
- Virtual Private Network (VPN) Servers
  - DMZs are commonly employed to host VPN servers, enabling secure remote access for external users without compromising the internal network's integrity.
- Intrusion Detection and Prevention
- Scalability and Flexibility
- Monitoring and Logging

- Advantages of DMZ are;
  - Enhanced Security
  - Provide Controlled Access
  - Scalability and Flexibility
  - Centralized Management and Monitoring
  - Support for Remote Access

- Disadvantages of DMZ are;
  - Initial Complex Implementation
  - Increased Maintenance Overhead
  - Higher Maintenance Cost
  - Limited Protection against Insider Threats
  - Requires Skilled Personnel

Intranet

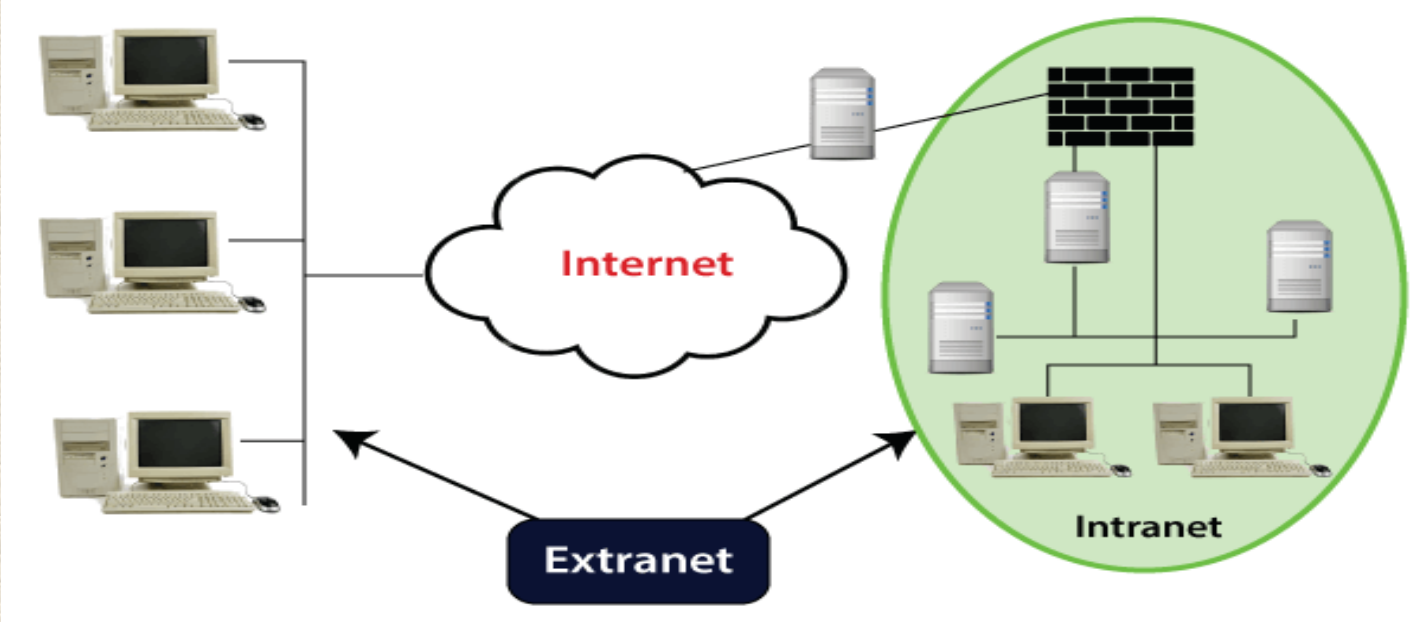
- The intranet is typically the network (or networks) that contains most of the organization's private resources, including computers, users, data, printers, and other network infrastructure equipment.
- An intranet is a private network within an organization that uses internet technologies and protocols to securely share information, collaboration tools, and computing resources among its members.
- It serves as an internal communication and information-sharing platform, fostering collaboration, knowledge-sharing, and efficient workflows within an organization.
- This is also known as the internal network, private network, local area network (LAN), trusted network, protected network, and company or organizational network.
- The basic figure are depicted below,



- Applications of Intranet;
  - Internal Communication
  - Knowledge Sharing
  - Employee Directory
  - Training and Development
  - Remote Work Support
  - Feedback and Surveys
  - Workflow Automation

Internet

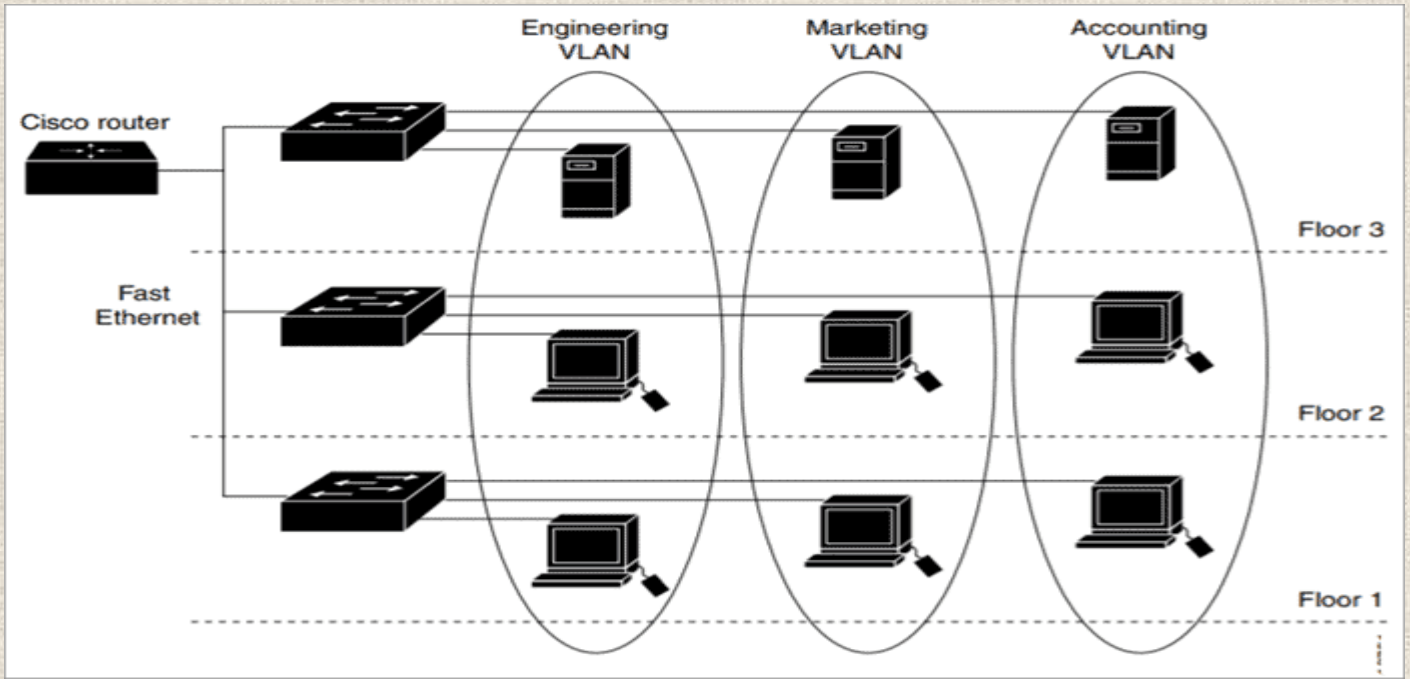
- The internet is a global network of interconnected computers and computer networks that use standardized communication protocols to exchange information.
- It is a vast and decentralized system that allows for the transfer of data, communication, and access to various resources across the globe.
- The purpose of an internet is to share information and technology between members of multiple organizations.
- Web browsers serve as gateways, allowing users to navigate websites hosted on servers worldwide, and WWW is a significant component of the Internet, providing a graphical interface for accessing information.
- The Basic Figure of Internet is given below;



- Applications of Internet;
  - Communication
  - Information Access
  - E-commerce
  - Education
  - Business
  - Entertainment
  - Social Networking
  - Research and Innovation
  - Remote Work

VLAN

- VLAN stands for Virtual Local Area Network, It is a networking technology that enables the logical segmentation of a physical network into multiple virtual networks.
- VLANs enable network administrators to group devices together, even if they are not physically connected to the same network switch, this grouping is done based on criteria such as department, team, or function, allowing for better network management, security, and efficiency.
- For example, in a large organization, different departments such as HR, Marketing, and IT may be assigned to separate VLANs.
- The basic figure of VLAN is given below, It is implemented using software and various networking devices.



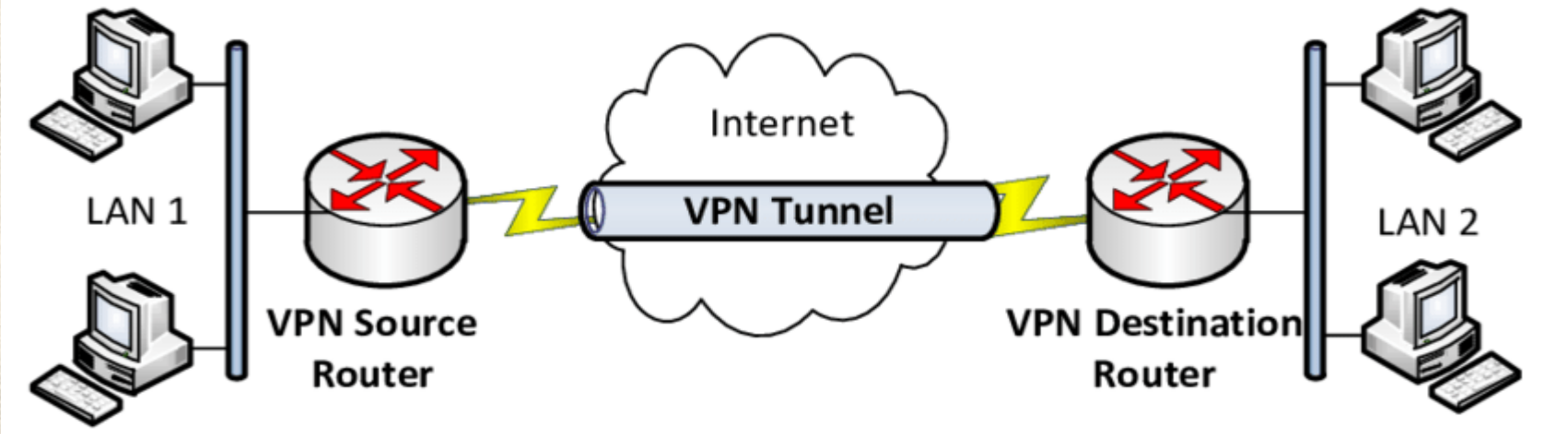
- Types of VLAN
  1. Port Based VLAN
     - This is the most common type of VLAN, where ports on a switch are assigned to different VLANs.
     - Devices connected to these ports are part of the same VLAN.
  2. MAC Based VLAN
     - VLAN membership is determined based on the MAC address of the device connected to the switch port.
  3. Protocol Based VLAN
     - VLAN membership is determined based on the protocol used in the network traffic.
     - For example, all IP traffic could be assigned to one VLAN, while all IPX traffic could be assigned to another.
  4. 802.1Q VLAN
     - This is a standard VLAN implementation that adds a VLAN tag to Ethernet frames, indicating the VLAN to which the frame belongs. This allows multiple VLANs to coexist on the same physical network.
  5. Voice VLAN
     - This type of VLAN is designed specifically for voice over IP (VoIP) traffic. It ensures that voice traffic is prioritized and can be easily segregated from other types of network traffic.

- Advantages of VLAN
  - Network Segmentation Facility
  - Improved Network Performance
  - Improved Security
  - Simplified Network Management
  - Simplified Network Troubleshooting
  - Flexibility and Scalability
- Disadvantages of VLAN
  - Setting up and managing VLANs can be complex.
  - Security Concern increase
  - Higher Cost
- Applications of VLAN
  - Departmental Segmentation
  - Video Conferencing
  - Remote Access
  - Guest Networks Creation
  - Virtualization Environments
  - Gaming Facility

- Security Implications of VLAN
  - Security implications refer to the potential consequences and impacts on the security of a system, network, or information due to certain actions, decisions, technologies, or vulnerabilities.
  - While VLANs provide advantages in terms of network segmentation, improved performance, and flexibility, it's essential to consider security implications to ensure the effectiveness of VLAN implementations.
  - Here are some security considerations and potential implications associated with VLANs
    - Management VLAN Security
    - Broadcast and Multicast Issues
    - Security in Shared VLANs
    - VLAN Infrastructure Security
    - Guest VLAN Security
    - Monitoring
    - Logging
    - Documentation
    - Change Control

Tunneling

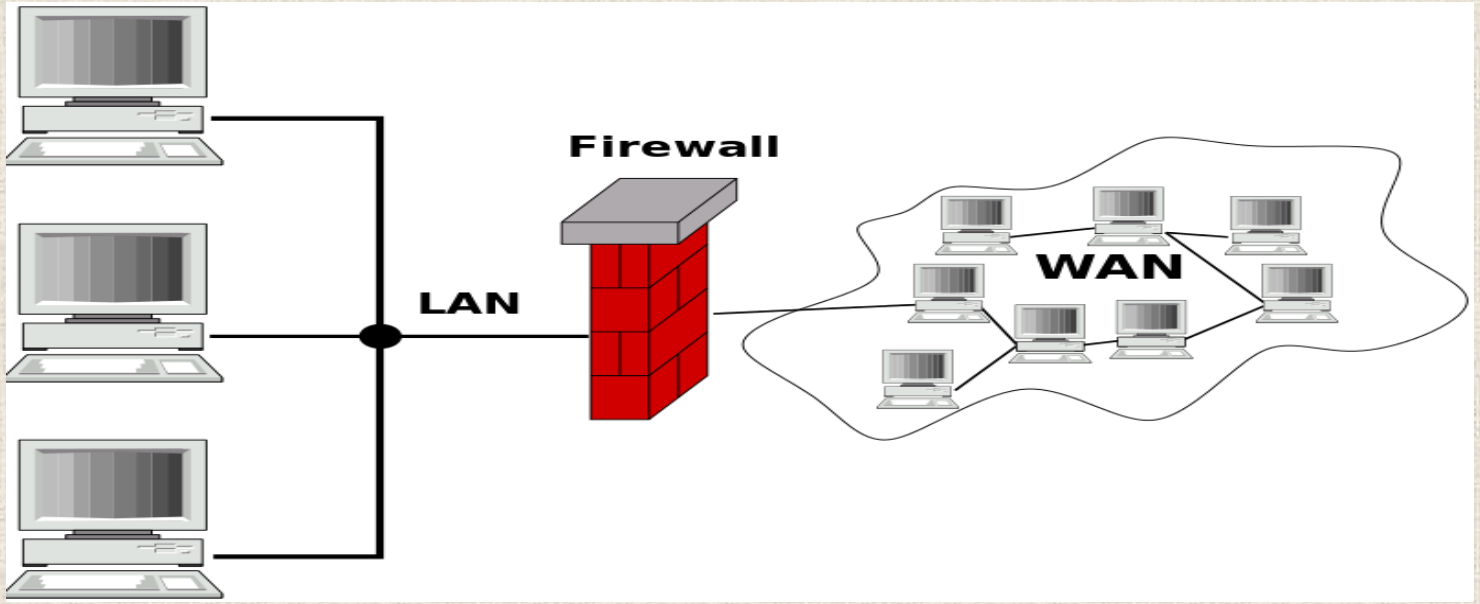
- Tunneling is a networking concept that involves encapsulating one network protocol within another, This process allows data from one network to be transported over another network, effectively creating a “Tunnel" through which the encapsulated data can travel.
- Tunneling is commonly used in various networking scenarios to enable communication between networks that may use different protocols or have different topologies.
- The best example of this concept VPN (virtual Private Network), VPN is a technology that enables secure and encrypted communication over the internet, allowing users to establish a private network connection over a public network.
- Why Needed: Assume that a company has a multiple branches and decides to use public internet to connect various branches.
- The basic figure is given below;



- Tunneling Protocols
  - Various protocols are used for tunneling, each with its own characteristics.
  - Some common tunneling protocols include:
    - Point-to-Point Tunneling Protocol (PPTP)
    - Layer 2 Tunneling Protocol (L2TP)
    - IPsec (Internet Protocol Security) Protocol
- Advantages of Tunneling
  - Provide Secure Connections
  - Helps to Implement VPN
  - Network Segmentation
  - Traffic Encapsulation
- Disadvantages of Tunneling
  - Performance Overhead
  - Complexity
  - Dependence on Protocols
  - Potential for Misuse
  - Resource Consumption

Firewall

- Firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules.
- The primary purpose of a firewall is to establish a barrier between a trusted internal network and untrusted external networks, such as the internet.
- Firewalls help prevent unauthorized access, monitor and control network communication, and enhance the overall security of a computer network.
- All messages entering or leaving the internet pass through the firewall, Which examines each message and blocks those that do not meet the specified security criteria.
- The basic place of firewall are depicted below;



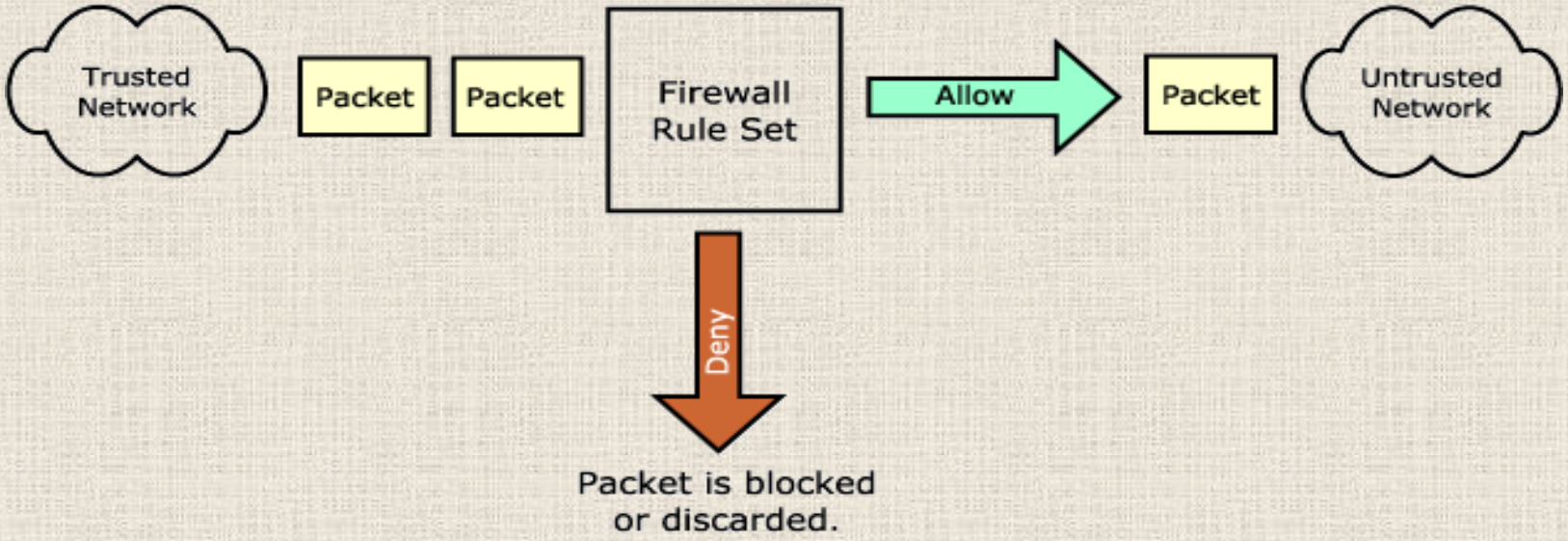
- Limitations of Firewall
  - The firewall cannot protect against attacks that bypass the firewall.
  - The firewall does not protect against internal threats.
  - A firewall does not prevent users from accessing malicious websites, which makes it vulnerable to internal threats and attacks.
  - The firewall cannot protect against the transfer of virus-infected programs or files.
  - The firewall is Expensive device.
- Design Goal of Firewall
  - All traffic from or to the internal network must pass through the firewall.
  - Only authorized traffic, as defined by the local security policy, will be allowed to pass.
  - The firewall itself is immune to penetration.

- Characteristics of Firewall
  - Service Control
    - ✓ Determines the types of Internet services that can be accessed.
  - Direction Control
    - ✓ Determines the direction in which particular service requests are allowed to flow through the firewall.
  - User Control
    - ✓ Controls access to a service according to which user is attempting to access it
  - Behavior Control
    - ✓ Controls how particular services are used.

- Need of Firewall (Advantages) (Application)
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Packet Filtering Firewall

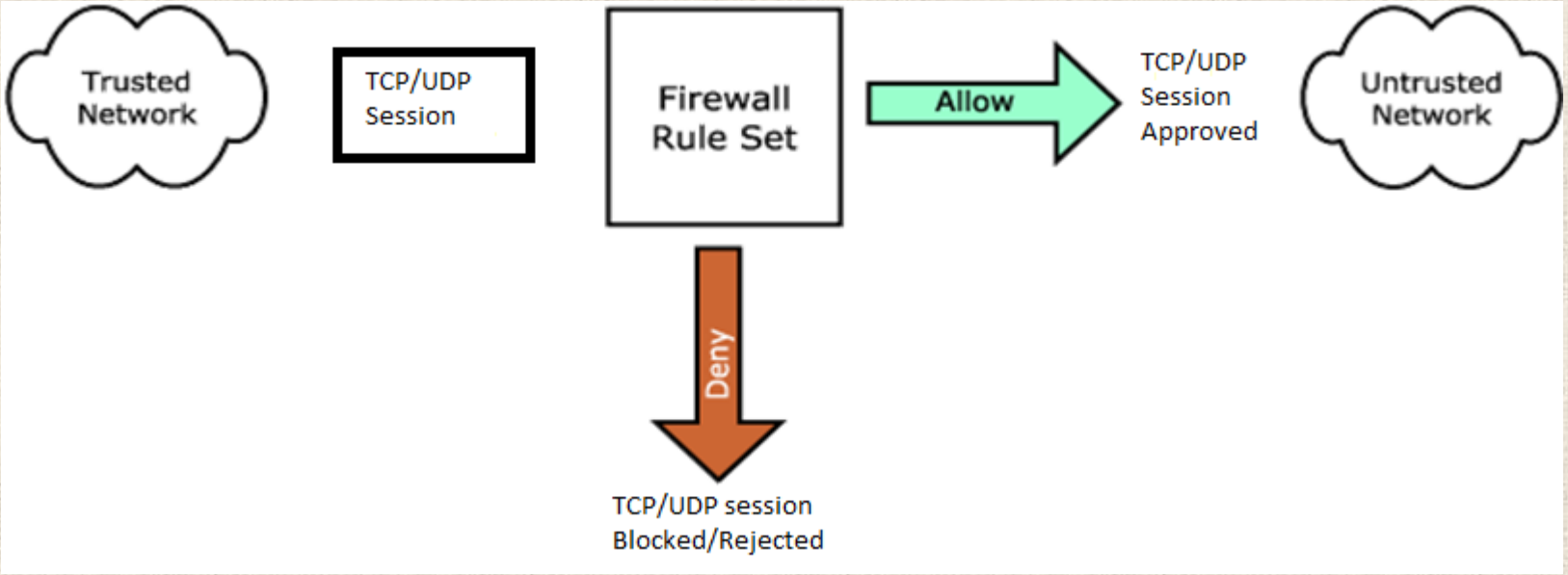
- A Packet-filtering firewall applies a set of rules to each incoming and outgoing IP packet and then forwards or discards the packet.
- Filtering rules are based on information contained in a network packet like Source IP address, Destination IP address, Source port, Destination port etc.
- As each packet passes through the firewall, it is examined and information contained in the header is compared to a pre-configured set of rules or filters.
- An allow or deny decision is made based on the results of the comparison.
- Operate at the Network Layer of the OSI model.
- The basic figure is given below;



- Advantages of Packet Filtering Firewall
  - Highly Efficient
  - Simple Architecture and Working
  - Higher Working Speed
  - Cost Effective
- Disadvantages of Packet Filtering Firewall
  - Less Effective due to Statelessness nature
  - Complex ACL (Access control lists) Management
- Applications of Packet Filtering Firewall
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Circuit-Level Gateway Firewall

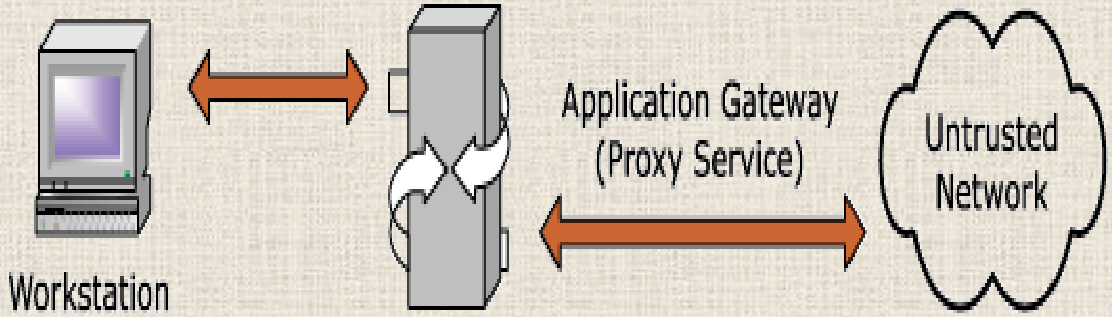
- Unlike a Packet filtering firewall, a circuit-level gateway does not examine individual packets. Instead, circuit-level gateways monitor TCP or UDP sessions.
- A Circuit level gateway firewall applies a set of rules to each incoming and outgoing TCP and UDP sessions and then forwards or discard it based on that rules.
- Operate at the Transport Layer of the OSI model.
- The basic figure is given below;



- Advantages of Circuit-Level Gateway Firewall
  - Ease of Setup and Management
  - Simple Design and Working
  - Cost Effective
  - Provide Effective Session Management
- Disadvantages of Circuit-Level Gateway Firewall
  - Absence of Application Layer Monitoring
  - Limited Content Inspection
  - Security Limitations
- Applications of Circuit-Level Gateway Firewall
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Application-Level Gateway Firewall

- An Application-level gateway firewall also known as a Proxy Server
- When a client issues a request from the untrusted network, a connection is established with the application gateway/proxy. The proxy determines if the request is valid (by comparing it to any rules or filters) and then sends a new request on behalf of the client to the destination.
- By using this method, a direct connection is never made from the trusted network to the untrusted network and the request appears to have originated from the application gateway/proxy.
- The response is sent back to the application gateway/proxy, which determines if it is valid and then sends it on to the client.
- Operate at the Application Layer of the OSI model.
- The basic figure is given below;



- Advantages of Application-Level Gateway Firewall
  - User Anonymity Protection
  - Enables Content Filtering
  - Enhanced Security Measures
- Disadvantages of Application-Level Gateway Firewall
  - Complex Configuration
  - Protocol Limitations
  - Higher Cost
  - Security Limitations
- Applications of Application-Level Gateway Firewall
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Stateful Inspection Firewall

- Stateful Inspection Firewall also known as Dynamic Packet Filtering Firewall, is a firewall technology that goes beyond the basic capabilities of traditional packet filtering firewalls.
- Stateful Inspection Firewalls operate at the Network Layer of the OSI model and provide a more advanced and context-aware approach to examining network traffic.
- These firewalls keep track of the state of active connections and make decisions based on the context of the entire communication session, offering improved security and awareness compared to stateless packet filtering.
- Features of it:
    - Connection Tracking
    - Context Awareness
    - Dynamic Rule Adjustment
    - Improved Security
    - Connection Logging
    - Application Layer Support

- Advantages of Stateful Inspection Firewall
  - Provide Stateful Tracking
  - Provide Adaptive Rule Management
  - Enhanced Security Measures
  - Simplicity in Rule Configuration
- Disadvantages of Stateful Inspection Firewall
  - Lower Speed
  - Higher Cost
- Applications of Stateful Inspection Firewall
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Next Generation Firewall

- Next Generation Firewall (NGFW) is a sophisticated network security device that combines traditional firewall capabilities with advanced security features such as intrusion prevention, application awareness and control, and user identity management.
- NGFWs are designed to provide enhanced security and granular control over network traffic, going beyond the capabilities of traditional firewalls.
- They operate at multiple layers of the OSI model, including the Network Layer and the Application Layer.
- Features of it:
    - Deep Packet Inspection
    - Application Awareness and Control
    - Intrusion Prevention System (IPS) Support
    - User Identity Management
    - SSL/TLS Support
    - Advanced Threat Protection
    - Cloud Integration
    - Integration with Security Services
    - Centralized Management and Reporting
    - Virtualization Support

- Advantages of Next Generation Firewall
  - Provide Proactive Threat Defense
  - Provide Adaptive Rule Management
  - Enhanced Security Measures
  - User Centric Application
- Disadvantages Next Generation Firewall
  - Integration Challenges
  - Complex Configuration
  - Higher Cost
- Applications Next Generation Firewall
  - Access Control
  - Network Security
  - Prevention of Unauthorized Access
  - Protection Against Cyber Attacks
  - Confidentiality and Privacy
  - Monitoring and Logging
  - Network Segmentation
  - Virtual Private Network (VPN) Support

Intruders

- The most common threat to security is an attack by an Intruder.
- Intruders are often referred to as hackers and are the most harmful factors contributing to the vulnerability of security, they have immense knowledge and an in-depth understanding of technology and security.
- Intruders crack the privacy of users and aim to steal the confidential information of the users, the stolen information is then sold to third parties, which aim at misusing the information for their own personal or professional gains.
- Three Classes of Intruders:
  1. Masquerader
     - An individual who is not authorized to use the computer and who penetrates a system's access controls to exploit a legitimate user's account.
  2. Misfeasor
     - A legitimate user who accesses data, programs, or resources for which such access is not authorized, or who is authorized for such access but misuses his or her privileges.
  3. Clandestine User
     - An individual who seizes supervisory control of the system and uses this control to escape auditing and access controls.

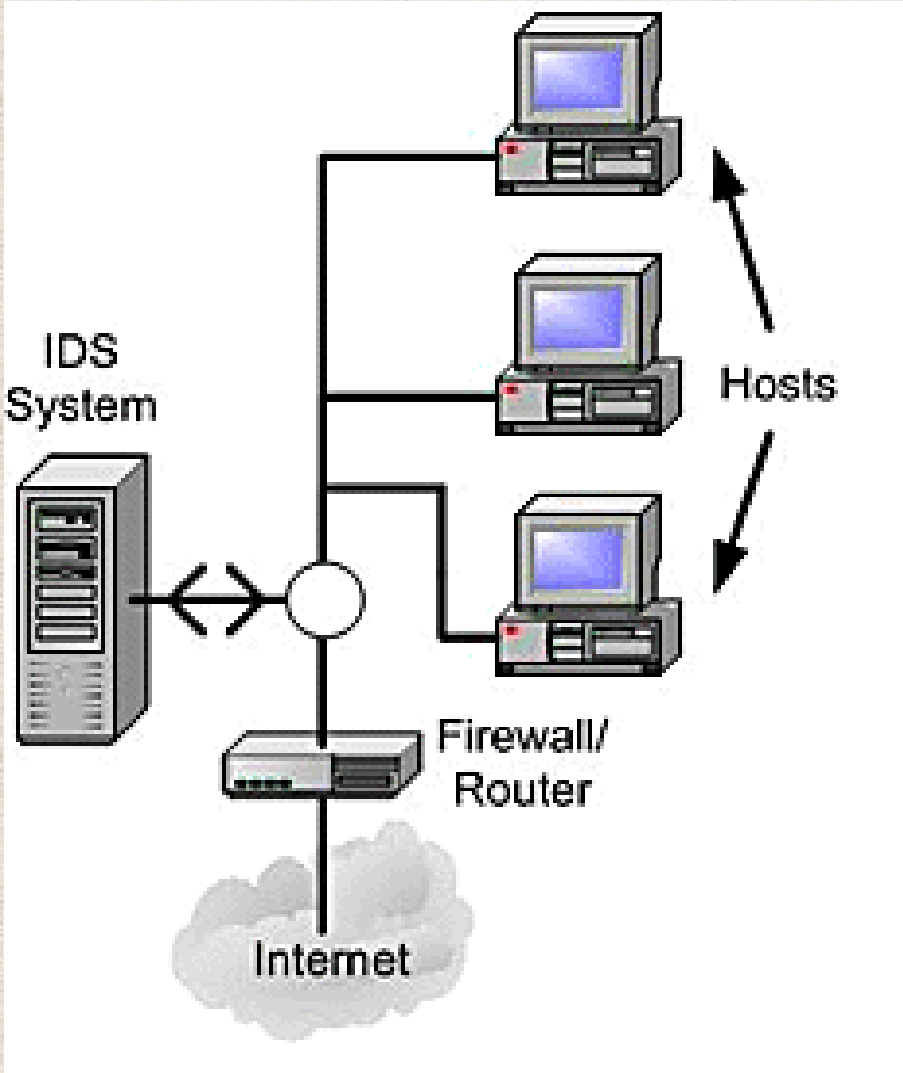
Intrusion Detection System (IDS)

- IDS is a device or software application designed to monitor and analyze network and system activities for signs of malicious behavior or security policy violations, after that it produces reports to a management station.
- IDS is often part of a broader security strategy, complementing other security measures such as firewalls, antivirus software, and access controls.
- Intrusion
  - A set of actions aimed to compromise the security services and principals.
- Intrusion Detection
  - The process of identifying and responding to intrusion activities.
- There are two main types of IDS:
  - Network-Based IDS (NIDS)
  - Host-Based IDS (HIDS).

- The Key Functions of IDS are listed below;
  - Monitoring
    - IDS continuously monitors and analyzes network and system activities.
  - Alerting
    - When suspicious or potentially malicious activity is detected, the IDS generates alerts to notify security administrators.
  - Logging
    - IDS systems maintain logs of detected events.
  - Response
    - While IDS itself doesn't prevent intrusions, it can trigger automated responses or alert security personnel to take appropriate actions.
  - Signature-Based Detection
    - Involves comparing observed activities against a database of known attack patterns or signatures.
  - Anomaly-Based Detection
    - Establishes a baseline of normal behavior and raises alerts when deviations from this baseline occur.
  - Heuristic-Based Detection
    - Utilizes rules and algorithms to identify new, previously unknown threats based on certain characteristics or behaviors.

Network Based IDS (NIDS)

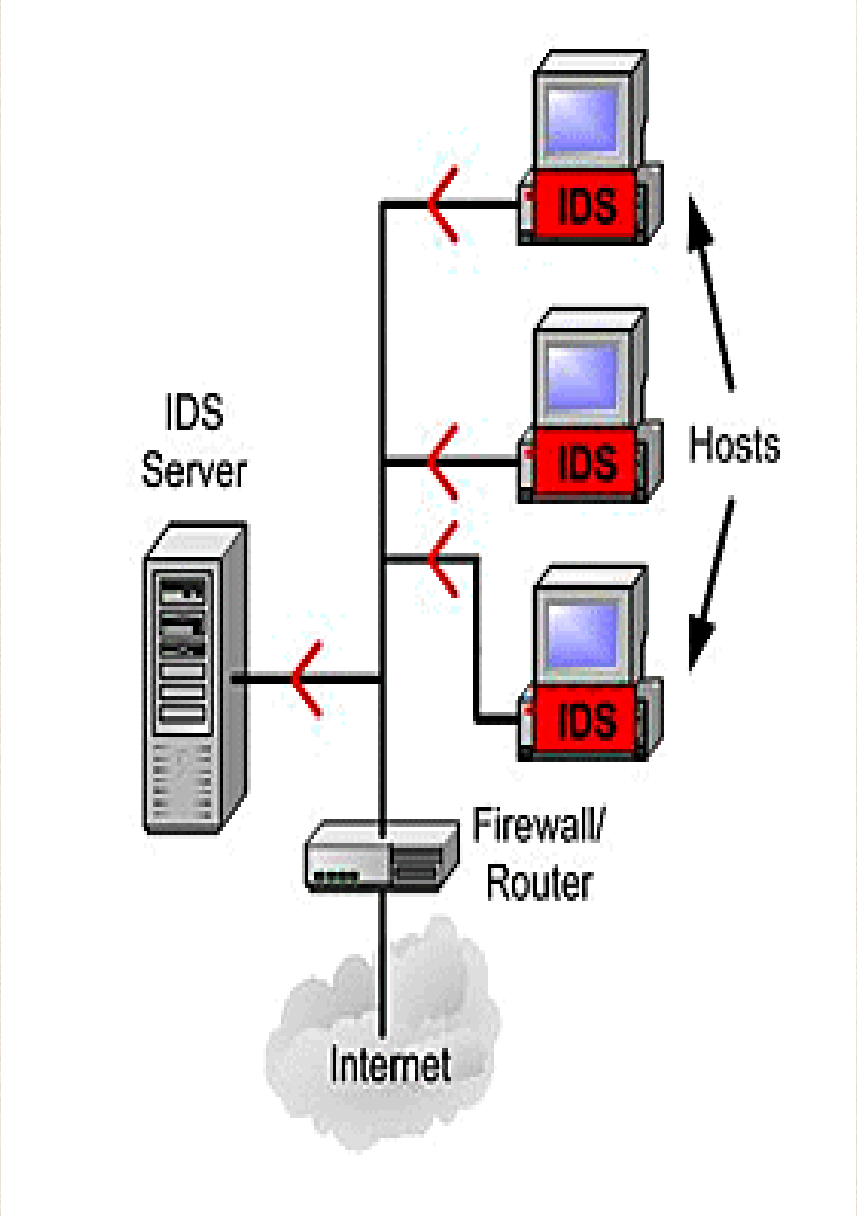
- NIDS monitors and analyzes network traffic to identify and respond to suspicious or malicious activities.
- NIDS is designed to operate at the network level, examining packets and traffic patterns to detect potential security threats.
- When threats are discovered, based on its severity, the system can take action such as notifying administrators, or barring the source IP address from accessing the network.
- The Basic Figure of NIDS are given below;



- Advantages of NIDS
  - Comprehensive Network Visibility
  - Centralized Monitoring
  - Early Threat Detection
  - Detects Known Attack Signatures
  - Scalability
  - Real Time Monitoring
  - Cost-Effective
- Disadvantages of NIDS
  - Difficulty in Analyzing Encrypted Traffic
  - Inability to Monitor Host-Level Activities
  - Additional Network Overhead
  - Limited Protection against Insider Threats
  - Complexity of Rule Configuration
  - Limited Response Capabilities
  - Dependency on Regular Updates

Host Based IDS (HIDS)

- HIDS monitors and analyzes the internals of a computing system, such as servers, workstations, or individual devices, for signs of malicious activity or security policy violations.
- Unlike Network-based IDS (NIDS), which focuses on monitoring network traffic, HIDS operates at the host level, examining activities on a specific device.
- If the critical system files were modified or deleted, the alert is sent to the administrator to investigate.
- The Basic Figure of HIDS are given below;



- Advantages of HIDS
  - Individual Host Protection
  - Insider Threat Detection
  - Customization for Specific Hosts
  - Detects Unauthorized Access
  - File Integrity Monitoring
  - Detects attacks that a network based IDS fail to detect.
  - Lower Cost
- Disadvantages of HIDS
  - Resource Consumption
  - Complexity in Managing Alerts
  - High Deployment and Maintenance Overhead
  - Dependency on Host Integrity
  - Difficulty in Scalability
  - Limited Network-Level Visibility
  - Inability to Monitor Encrypted Traffic

Important Questions

Unit-4 (CO-4)	
Sr No.	**Question**
1	Explain DMZ with Suitable Diagram. OR Describe Demilitarized Zone (DMZ) and its applications.
2	Explain Internet with Suitable Diagram.
3	Explain Intranet with Suitable Diagram.
4	Explain VLAN with Suitable Diagram,Also Explain any Three Types of It. OR Describe Virtual LAN (VLAN) security topology
5	Explain Tunneling with Suitable Diagram.
6	Define Firewall, List Needs (Application) of Firewall. OR Describe need and working of firewall. OR Describe the working of the firewall in detail with a diagram. OR List types of firewalls and write about needs of firewall OR Explain Packet Filtering Firewall. OR Explain Circuit-Level Gateway Firewall. OR Explain Application-Level Gateway Firewall. OR Explain Stateful Inspection Firewall. OR Explain Next Generation Firewall.
7	Define IDS,List Key Functions of IDS. OR Describe working and components of IDS. OR Explain NIDS with Suitable Diagram, Also List Advantages and Disadvantages. OR Explain HIDS with Suitable Diagram, Also List Advantages and Disadvantages.